



Vardan EnviroLab

Laboratory: Plot No. 82A, Sector - 5, IMT Manesar, Gurugram - 122051 (Haryana)
NABL Accredited | MoEF&CC Recognized | ISO 9001 | ISO 14001 | ISO 45001

Test Report

Sample Number: VEL/HCL/A/03
Name & Address of the Party: M/s Hindustan Copper Limited,
Indian Copper Complex,
Ghatsila, Disst.- Singhbhum,
Jharkhand India.

Report No.: VEL/A/2106/24/003
Format No.: 7.8 F-01
Party Reference No.: NIL
Reporting Date: 29/06/2021
Period of Analysis: 24/06/2021 to 29/06/2021
Receipt Date: 24/06/2021

Sample Description: AMBIENT AIR QUALITY MONITORING

General Information:-

Sample collected by : Vardan EnviroLab Representative (Mr. Kamal)
Sampling Location : Near Admin Building Surda
Instrument Used : RDS & FPS Sampler with all Accessories
Instrument Code : VEL/RDS/10 & VEL/FPS/10
Instrument Calibration Status : Calibrated
Meteorological condition during monitoring : Rain Fall
Date of Monitoring : 15/06/2021 to 16/06/2021
Time of Monitoring : 10:30 AM – 10:30 AM
Ambient Temperature (°C) : Min.27°C & Max. 30°C
Surrounding Activity : Human, Vehicular & Construction Activities
Sampling & Analysis Protocol : IS:5182 & CPCB Guidelines
Parameter Required : As per Work Order

S. No.	Parameter	Protocol	Result	Unit	NAAQS* Limit
1.	Particulate Matter (PM _{2.5})	"SOP No. VEL/SOP/01, Section No. SP 63	37.52	µg/m ³	60
2.	Particulate Matter (PM ₁₀)	IS: 5182 (P-23), Gravimetric Method	68.61	µg/m ³	100
3.	Nitrogen Dioxide (NO _x)	IS: 5182 (P-6), Jacob & Hochheiser	15.47	µg/m ³	80
4.	Sulphur Dioxide (SO ₂)	IS: 5182 (P-2), Modified West and Gaeke	8.24	µg/m ³	80

* NAAQS - National Ambient Air Quality Standards; Schedule-VII, [Rule 3 (3B)], [Part-II-sec.-3(i)] 18.11.2009. #SOP- Laboratory Standard Operating Procedure
Remark:-All parameters are under limits as per CPCB NAAQS 2009. Instruments Used - Respirable Dust Sampler (APM 460), Calibration Date- 17/05/21,
Fine Particulate Sampler (EI 133), Calibration Date- 17/05/21

KOMAL SINGH
ANALYST
(Tested By)

(Checked By)



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Test Report

Sample Number: VEL/HCL/A/04
Name & Address of the Party: M/s Hindustan Copper Limited,
Indian Copper Complex,
Ghatsila, Disst.- Singhbhum,
Jharkhand India.

Report No.: VEL/A/2106/24/004
Format No.: 7.8 F-01
Party Reference No.: NIL
Reporting Date: 29/06/2021
Period of Analysis: 24/06/2021 to 29/06/2021
Receipt Date: 24/06/2021

Sample Description: AMBIENT AIR QUALITY MONITORING

General Information:-

Sample collected by : Vardan EnviroLab Representative (Mr. Kamal)
Sampling Location : Surda 3 # Shaft North Side
Instrument Used : RDS & FPS Sampler with all Accessories
Instrument Code : VEL/RDS/10 & VEL/FPS/10
Instrument Calibration Status : Calibrated
Meteorological condition during monitoring : Rain Fall
Date of Monitoring : 16/06/2021 to 17/06/2021
Time of Monitoring : 11:36 AM – 11:36 AM
Ambient Temperature (°C) : Min.23°C & Max. 27°C
Surrounding Activity : Human, Vehicular & Construction Activities
Sampling & Analysis Protocol : IS:5182 & CPCB Guidelines
Parameter Required : As per Work Order

S. No.	Parameter	Protocol	Result	Unit	NAAQS ^W Limit
1.	Particulate Matter (PM _{2.5})	#SOP No. VEL/SOP/01, Section No. SP 63	39.84	µg/m ³	60
2.	Particulate Matter (PM ₁₀)	IS: 5182 (P-23), Gravimetric Method	70.14	µg/m ³	100
3.	Nitrogen Dioxide (NO _x)	IS: 5182 (P-6), Jacob & Hochheiser	21.58	µg/m ³	80
4.	Sulphur Dioxide (SO ₂)	IS: 5182 (P-2), Modified West and Gaeke	7.56	µg/m ³	80

* NAAQS – National Ambient Air Quality Standards; Schedule-VII, [Rule 3 (3B)], [Part-II-sec.-3(i)] 18.11.2009. #SOP- Laboratory Standard Operating Procedure

Remark:- All parameters are under limits as per CPCB NAAQS 2009. Instruments Used – Respirable Dust Sampler (APM 460), Calibration Date- 17/05/21,
Fine Particulate Sampler (EI 133), Calibration Date- 17/05/21

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Test Report

Sample Number: VEL/HCL/A/05
Name & Address of the Party: M/s Hindustan Copper Limited,
Indian Copper Complex,
Ghatsila, Disst.- Singhbhum,
Jharkhand India.

Report No.: VEL/A/2106/24/005
Format No.: 7.8 F-01
Party Reference No.: NIL
Reporting Date: 29/06/2021
Period of Analysis: 24/06/2021 to 29/06/2021
Receipt Date: 24/06/2021

Sample Description: FUGITIVE AIR QUALITY MONITORING

General Information:-

Sample collected by : Vardan EnviroLab Representative (Mr. Kamal)
Sampling Location : Near Mines Manager Office 4 # Shaft South Side Surda
Instrument Used : RDS & FPS Sampler with all Accessories
Instrument Code : VEL/RDS/09 & VEL/FPS/09
Instrument Calibration Status : Calibrated
Meteorological condition during monitoring : Clear Sky
Date of Monitoring : 18/06/2021
Time of Monitoring : 10:02 AM – 06:02 PM
Ambient Temperature (°C) : Min.25°C & Max. 31°C
Surrounding Activity : Human, Vehicular & Construction Activities
Sampling & Analysis Protocol : IS:5182 & CPCB Guidelines
Parameter Required : As per Work Order

S. No.	Parameter	Protocol	Result	Unit	Limits As per factory Act 1948, Schedule-II (Sec-41F)
1.	Particulate Matter (PM _{2.5})	#SOP No. VEL/SOP/01, Section No. SP 63	0.059	mg/m ³	--
2.	Particulate Matter (PM ₁₀)	IS: 5182 (P-23), Gravimetric Method	0.099	mg/m ³	--
3.	Nitrogen Dioxide (NO _x)	IS: 5182 (P-6), Jacob & Hochheiser	0.035	mg/m ³	6
4.	Sulphur Dioxide (SO ₂)	IS: 5182 (P-2), Modified West and Gaeke	0.029	mg/m ³	5

Remark:- All parameters are under limits as per Factory Act 1948, Schedule-II (Sec 41 F). Instruments used – Respirable Dust Sampler (APM460),
Calibration Date-17/05/21, Fine Particulate Sampler (EI 133), Calibration Date- 17/05/21

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Test Report

Sample Number: VEL/HCL/A/06
Name & Address of the Party: M/s Hindustan Copper Limited,
Indian Copper Complex,
Ghatsila, Disst.- Singhbhum,
Jharkhand India.

Report No.: VEL/A/2106/24/006
Format No.: 7.8 F-01
Party Reference No.: NIL
Reporting Date: 29/06/2021
Period of Analysis: 24/06/2021 to 29/06/2021
Receipt Date: 24/06/2021

Sample Description: FUGITIVE AIR QUALITY MONITORING

General Information:-

Sample collected by : Vardan EnviroLab Representative (Mr. Kamal)
Sampling Location : Surda 3 # Shaft Under Ground 5 Level
Instrument Used : RDS & FPS Sampler with all Accessories
Instrument Code : VEL/RDS/09 & VEL/FPS/09
Instrument Calibration Status : Calibrated
Meteorological condition during monitoring : Clear Sky
Date of Monitoring : 17/06/2021
Time of Monitoring : 12:35 PM – 08:35 PM
Ambient Temperature (°C) : Min.24°C & Max. 28°C
Surrounding Activity : Human, Vehicular & Construction Activities
Sampling & Analysis Protocol : IS:5182 & CPCB Guidelines
Parameter Required : As per Work Order

S. No.	Parameter	Protocol	Result	Unit	Limits As per factory Act 1948, Schedule-II (Sec-41F)
1.	Particulate Matter (PM _{2.5})	#SOP No. VEL/SOP/01, Section No. SP 63	0.052	mg/m ³	--
2.	Particulate Matter (PM ₁₀)	IS: 5182 (P-23), Gravimetric Method	0.063	mg/m ³	--
3.	Nitrogen Dioxide (NO _x)	IS: 5182 (P-6), Jacob & Hochheiser	0.039	mg/m ³	6
4.	Sulphur Dioxide (SO ₂)	IS: 5182 (P-2), Modified West and Gaeke	0.031	mg/m ³	5

Remark:-All parameters are under limits as per Factory Act 1948, Schedule-II (Sec 41 F). Instruments used – Respirable Dust Sampler (APM460)
Calibration Date-17/05/21, Fine Particulate Sampler (EI 133), Calibration Date- 17/05/21

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Test Report

Sample Number:	VEL/HCL/AN/03	Report No.:	VEL/AN/2106/24/003
Name & Address of the Party:	M/s Hindustan Copper Limited, Indian Copper Complex, Ghatsila, Disst.- Singhbhum, Jharkhand India.	Format No.:	7.8 F-01
		Party Reference No.:	NIL
		Reporting Date:	29/06/2021
		Receipt Date:	24/06/2021

Sample Description: AMBIENT NOISE LEVEL MONITORING

General Information:-

Sample collected by	: Vardan EnviroLab Representative (Mr. Kamal)
Sampling Location	: Near Admin Building Surda
Instrument Used	: Sound Level Meter
Instrument Calibration Status	: Calibrated
Meteorological condition during monitoring	: Rain Fall
Date of Monitoring	: 15/06/2021 to 16/06/2021
Time of Monitoring	: 06:00 AM to 06:00 AM
Ambient Temperature (°C)	: Min.27°C & Max. 30°C
Surrounding Activity	: Human, Vehicular, Plant & Other Activities
Sampling & Analysis Protocol	: IS:9989
Sampling Duration	: 24 Hours
Parameter Required	: As per Work Order

S. No.	Parameters	Protocol	Test Result dB (A)		Unit
			Day Time (06:00 am to 10:00 pm)	Night Time (10:00 pm to 06:00 am)	
1.	L_{max}	IS 9989	76.9	64.2	dB (A)
2.	L_{min}	IS 9989	58.5	51.7	dB (A)
3.	L_{eq}	IS 9989	68.7	58.3	dB (A)
4.	CPCB Limits in dB(A) L_{eq} (Industrial Area)	-	75.00	70.00	dB(A)

Note: *A "decibel" is a unit in which noise is measured.

Remark: All parameters are under limits as per CPCB Noise Standards 2000.

Instruments Used – Noise Meter (Q657612), Calibration Date- 17/05/21.

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Test Report

Sample Number:	VEL/HCL/AN/04	Report No.:	VEL/AN/2106/24/004
Name & Address of the Party:	M/s Hindustan Copper Limited, Indian Copper Complex, Ghatsila, Disst.- Singhbhum, Jharkhand India.	Format No.:	7.8 F-01
		Party Reference No.:	NIL
		Reporting Date:	29/06/2021
		Receipt Date:	24/06/2021

Sample Description: AMBIENT NOISE LEVEL MONITORING

General Information:-

Sample collected by	: Vardan EnviroLab Representative (Mr. Kamal)
Sampling Location	: Surda 3 # Shaft North Side
Instrument Used	: Sound Level Meter
Instrument Calibration Status	: Calibrated
Meteorological condition during monitoring	: Rain Fall
Date of Monitoring	: 16/06/2021 to 17/06/2021
Time of Monitoring	: 06:00 AM to 06:00AM
Ambient Temperature (°C)	: Min.23°C & Max. 27°C
Surrounding Activity	: Human, Vehicular, Plant & Other Activities
Sampling & Analysis Protocol	: IS:9989
Sampling Duration	: 24 Hours
Parameter Required	: As per Work Order

S. No.	Parameters	Protocol	Test Result dB (A)		Unit
			Day Time (06:00 am to 10:00 pm)	Night Time (10:00 pm to 06:00 am)	
1.	L _{max}	IS 9989	71.4	58.9	dB (A)
2.	L _{min}	IS 9989	48.3	42.7	dB (A)
3.	L _{eq}	IS 9989	61.8	49.5	dB (A)
4.	CPCB Limits in dB(A) [*] Leq (Industrial Area)	--	75.00	70.00	dB(A)

Note: *A "decibel" is a unit in which noise is measured.

Remark: All parameters are under limits as per CPCB Noise Standards 2000.

Instruments Used - Noise Meter (Q657612), Calibration Date- 17/05/21.

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Test Report

Sample Number:	VEL/HCL/W/01	Report No.:	VEL/W/2106/24/001
Name & Address of the Party:	M/s Hindustan Copper Limited, Indian Copper Complex, Ghatsila, Disst.- Singhbhum, Jharkhand India.	Format No.:	7.8 F-01
Sample Description:	Drinking Water Sample	Party Reference No.:	NIL
Sampling Location:	Canteen Potability Water (Surda Mine)	Reporting Date:	29/06/2021
Sample Collected By:	VEL Representative (Mr. Kamal)	Period of Analysis :	24/06/2021 to 29/06/2021
Sampling & Analysis Protocol:	APHA, 23 rd Edition 2017	Receipt Date:	24/06/2021
Parameter Required:	As per Work Order	Sampling Date:	21/06/2021
		Sampling Type:	Grab
		Sampling Quantity:	2.0 Ltr. + 200 ml
		Preservation:	Refrigerated

S. No.	Parameter	Test Method	Result	Unit	Limits of IS:10500 -2012	
					Requirement (Acceptable Limit)	Permissible limit in the Absence of Alternate Source
1.	pH (at 25°C)	APHA, 4500 H ⁺ B Electrometric Method	7.35	--	6.5 to 8.5	No Relaxation
2.	Colour	APHA, 2120 B, Visual Comparison Method	*BDL (**DL 5Hazen)	Hazen	5	15
3.	Turbidity	APHA, 2130 B, Nephelometric Method	*BDL (**DL-1.0 NTU)	NTU	1	5
4.	Odour	APHA, 2150 B, Threshold Odour Method	Agreeable	--	Agreeable	Agreeable
5.	Taste	APHA, 2160 B, Flavor Threshold Test Method	Agreeable	--	Agreeable	Agreeable
6.	Total Hardness as CaCO ₃	APHA, 2340 C, EDTA Titrimetric Method	165.00	mg/l	200	600
7.	Calcium as Ca	APHA, 3500 Ca B, EDTA Titrimetric Method	54.26	mg/l	75	200
8.	Alkalinity as CaCO ₃	APHA, 2320 B, Titration Method	151.2	mg/l	200	600
9.	Chloride as Cl	APHA, 4500 ClB, Argentometric Method	39.7	mg/l	250	1000
10.	Cyanide as CN	APHA, 4500 CND Titrimetric Method	*BDL(**DL-0.02 mg/l)	mg/l	0.05	No Relaxation
11.	Magnesium as Mg	APHA, 3500 Mg B, Calculation Method	7.30	mg/l	30	100
12.	Total Dissolved Solids	APHA, 2540 C, Gravimetric Method	315.00	mg/l	500	2000
13.	Sulphate as SO ₄	APHA, 4500 E, Turbidimetric Method	12.56	mg/l	200	400
14.	Fluoride as F	APHA, 4500 F DSponds Method	0.22	mg/l	1.0	1.5
15.	Nitrate as NO ₃	IS 3025 (P-34), Chromotropic Method	1.33	mg/l	45	No Relaxation
16.	Iron as Fe	APHA, 3500 Fe B 1,10 Phenanthroline Method	0.16	mg/l	1.0	No relaxation
17.	Aluminium as Al	APHA, 3111 D, Direct Nitrous Oxide, Acetylene Flame Method	*BDL(**DL-0.002 mg/l)	mg/l	0.03	0.2
18.	Boron as B	APHA, 4500 C, Carmine Method	*BDL(**DL-0.01 mg/l)	mg/l	0.5	2.4
19.	Total Chromium as Cr	APHA, 3111 B, Direct Acetylene Flame Method	*BDL(**DL-0.01 mg/l)	mg/l	0.05	No Relaxation

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Test Report

Sample No.:VEL/HCL/W/01				Report No:VEL/W/2106/24/001		
S.No	Parameter	Test Method	Result	Unit	Limits of IS:10500-2012	
					Requirement (Acceptable) Limit	Permissible limit in the Absence of Alternate Source
20.	Phenolic Compounds	APHA, 5530 C, Chloroform Extraction Method	*BDL(**DL-0.0004 mg/l)	mg/l	0.001	0.002
21.	[#] Mineral Oil	APHA, 5520 C, Partition Infra Red	*BDL(**DL-0.05 mg/l)	mg/l	0.5	No Relaxation
22.	[#] Anionic Detergents as MBAS	APHA, 5540 C, MBAS Method	*BDL(**DL-0.05 mg/l)	mg/l	0.2	1.0
23.	Zinc as Zn	APHA, 3111 B, Direct Acetylene Flame Method	*BDL(**DL-0.05 mg/l)	mg/l	5	15
24.	Copper as Cu	APHA, 3111 B, Direct Acetylene Flame Method	*BDL(**DL-0.031 mg/l)	mg/l	0.05	1.5
25.	Manganese as Mn	APHA, 3111 B, Direct Acetylene Flame Method	*BDL(**DL-0.01 mg/l)	mg/l	0.1	0.3
26.	Cadmium as Cd	APHA, 3111 B, Direct Acetylene Flame Method	*BDL(**DL-0.001 mg/l)	mg/l	0.003	No Relaxation
27.	Lead as Pb	APHA, 3111 B, Direct Acetylene Flame Method	*BDL(**DL-0.005 mg/l)	mg/l	0.01	No Relaxation
28.	Selenium as Se	APHA, 3114 B, AAS Method	*BDL(**DL-0.005 mg/l)	mg/l	0.01	No Relaxation
29.	Arsenic as As	APHA, 3114 B, AAS Method	*BDL(**DL-0.00 mg/l)	mg/l	0.01	No Relaxation
30.	Mercury as Hg	APHA, 3112 B, Cold Vapor AAS Method	*BDL(**DL-0.0005 mg/l)	mg/l	0.001	No Relaxation
31.	Total Coliform	IS: 15185:2002 (RA-2016)	Absent	cfu/100ml	Shall not be detectable in any 100 ml sample	
32.	E. Coli	IS: 15185:2002 (RA-2016)	Absent	cfu/100ml	Shall not be detectable in any 100 ml sample	

Note:- *BDL-Below Detection Limit, **DL- Detection Limit

[#]These parameter are not covered in our NABL Scope.

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Test Report

Sample Number:	VEL/HCL/W/03	Report No.:	VEL/W/2106/24/003
Name & Address of the Party:	M/s Hindustan Copper Limited, Indian Copper Complex, Ghatsila, Disst.- Singhbhum, Jharkhand India.	Format No.:	7.8 F-01
Sample Description:	Ground Water Sample	Party Reference No.:	NIL
Sampling Location:	Durga Basti Hand Pump Water (Surda Mine)	Reporting Date:	29/06/2021
Sample Collected By:	VEL Representative (Mr. Kamal)	Period of Analysis:	24/06/2021 to 29/06/2021
Sampling & Analysis Protocol:	IS 3025 & APHA, 23 rd Edition 2017	Receipt Date:	24/06/2021
Parameter Required:	As Per Work Order	Sampling Date:	21/06/2021
		Sampling Quantity:	Grab
		Sampling Type:	2.0 Ltr. + 200 ml
		Preservation:	Refrigerated

S. No.	Parameter	Test Method	Result	Unit	Limits of IS:10500 -2012	
					Requirement (Acceptable Limit)	Permissible limit in the Absence of Alternate Source
1.	pH (at 25°C)	APHA, 4500 H ⁺ B Electrometric Method	6.59	--	6.5 to 8.5	No Relaxation
2.	Colour	APHA, 2120 B, Visual Comparison Method	*BDL (**DL 1.0 Hazen)	Hazen	5	15
3.	Turbidity	APHA, 2130 B, Nephelometric Method	*BDL (**DL 1.0 NTU)	NTU	1	5
4.	Total Hardness as CaCO ₃	APHA, 2340 C, EDTA Titrimetric Method	296.00	mg/l	200	600
5.	Calcium as Ca	APHA, 3500 Ca B, EDTA Titrimetric Method	72.56	mg/l	75	200
6.	Alkalinity as CaCO ₃	APHA, 2320 B, Titration Method	242.21	mg/l	200	600
7.	Chloride as Cl	APHA, 4500 ClB, Argentometric Method	49.85	mg/l	250	1000
8.	Magnesium as Mg	APHA, 3500 Mg B, Calculation Method	27.90	mg/l	30	100
9.	Total Suspended solids	APHA, 2540 C, Gravimetric Method	3.56	mg/l	--	--
10.	Sulphate as SO ₄	APHA, 4500 E, Turbidimetric Method	20.14	mg/l	200	400
11.	Fluoride as F	APHA, 4500 F D Spands Method	0.42	mg/l	1.0	1.5
12.	BOD (3 Days at 27°C)	APHA, 5210 C/ IS: 3025 (Part-44)	*BDL (**DL-2.0 mg/l)	mg/l	--	--
13.	COD	APHA, 5220 B, Open Reflux Method	*BDL (**DL-4.0 mg/l)	mg/l	--	--
14.	Nitrate as NO ₃	IS:3025 (Part-34), Chromotropic Method	8.52	mg/l	45	No Relaxation
15.	Iron as Fe	APHA, 3500 Fe B 1,10 Phenanthroline Method	0.24	mg/l	1.0	No relaxation
16.	Copper as Cu	APHA, 3111 B, Direct Acetylene Flame Method	0.40	mg/l	0.05	1.5

Remark:- Above test results confirm that this water is fit for drinking purpose as almost all the tested parameters are well within Acceptable limits.

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Test Report

Sample Number:	VEL/HCL/W/06	Report No.:	VEL/W/2106/24/006
Name & Address of the Party:	M/s Hindustan Copper Limited, Indian Copper Complex, Ghatsila, Disst.- Singhbhum, Jharkhand India.	Format No.:	7.8 F-01
Sample Description:	Surface Water Sample	Party Reference No.:	NIL
Sampling Location:	Surda Nala After ML Area (Surda)	Reporting Date:	29/06/2021
Sample Collected By:	VEL Representative (Mr. Kamal)	Period of Analysis:	24/06/2021 to 29/06/2021
Sampling & Analysis Protocol:	IS 3025 & APHA, 23 rd Edition 2017	Receipt Date:	24/06/2021
Parameter Required:	As Per Work Order	Sampling Date:	21/06/2021
		Sampling Quantity:	Grab
		Sampling Type:	2.0 Ltr. + 200 ml
		Preservation:	Refrigerated

S. No.	Parameter	Test Method	Result	Unit	Surface Water Quality Standard as per IS:2296 (Class C)
1.	pH (at 25°C)	APHA, 4500 H ⁺ B Electrometric Method	7.52	--	6.5-8.5
2.	Colour	APHA, 2120 B, Visual Comparison Method	*BDL (**DL 5Hazen)	Hazen	300
3.	Turbidity	APHA, 2130 B, Nephelometric Method	1.9	NTU	--
4.	Total Hardness as CaCO ₃	APHA, 2340 C, EDTA Titrimetric Method	112.00	mg/l	--
5.	Calcium as Ca	APHA, 3500 Ca B, EDTA Titrimetric Method	34.26	mg/l	--
6.	Alkalinity as CaCO ₃	APHA, 2320 B, Titration Method	95.14	mg/l	--
7.	Chloride as Cl	APHA, 4500 CIB, Argentometric Method	23.47	mg/l	600
8.	Magnesium as Mg	APHA, 3500 Mg B, Calculation Method	6.42	mg/l	--
9.	Total Suspended solids	APHA, 2540 C, Gravimetric Method	3.21	mg/l	--
10.	Sulphate as SO ₄	APHA, 4500 E, Turbidimetric Method	18.52	mg/l	400
11.	Fluoride as F	APHA, 4500 F D Spands Method	0.35	mg/l	1.5
12.	BOD (3 Days at 27°C)	APHA, 5210 C/ IS: 3025 (Part-44)	*BDL (**DL 2.0 mg/l)	mg/l	3
13.	COD	APHA, 5220 B, Open Reflux Method	7.26	mg/l	--
14.	Nitrate as NO ₃	IS:3025 (Part-34). Chromotropic Method	9.24	mg/l	50
15.	Iron as Fe	APHA, 3500 Fe B 1,10 Phenanthroline Method	0.21	mg/l	50
16.	Copper as Cu	APHA, 3111 B, Direct Acetylene Flame Method	0.57	mg/l	1.5

Remark:-All parameters are well within limits as per Surface Water Quality Standard as per IS: 2296, Class 'C', this water may use for general purpose.

(Tested By)

ARJUN RAWAT
(Checked By)
[Signature]



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Vardan EnviroLab

Laboratory: Plot No. 82A, Sector - 5, IMT Manesar, Gurugram - 122051 (Haryana)
NABL Accredited | MoEF&CC Recognized | ISO 9001 | ISO 14001 | ISO 45001

Test Report

Sample Number:	VEL/HCL/W/11	Report No.:	VEL/W/2106/24/011
Name & Address of the Party:	M/s Hindustan Copper Limited, Indian Copper Complex, Ghatsila, Distt.- Singhbhum, Jharkhand India.	Format No.:	7.8 F-01
Sample Description:	Recycle Water Sample	Party Reference No.:	NIL
Sampling Location:	Storage Tank – I (Surda Mines)	Reporting Date:	29/06/2021
Sample Collected By:	VEL Representative (Mr. Kamal)	Period of Analysis:	24/06/2021 to 29/06/2021
Sampling & Analysis Protocol:	IS: 10500-2012 & APHA	Receipt Date:	24/06/2021
Parameter Required:	As Per Work Order	Sampling Date:	21/06/2021
		Sampling Quantity:	2.0 Ltr.
		Sampling Type:	Grab
		Preservation:	Refrigerated

S. No.	Parameter	Test Method	Result	Unit
1.	pH (at 25°C)	APHA, 4500 H ⁺ B Electrometric Method	7.11	--
2.	Colour	APHA, 2120 B, Visual Comparison Method	*BDL (**DL 5 Hazen)	Hazen
3.	Turbidity	APHA, 2130 B, Nephelometric Method	1.0	NTU
4.	Total Hardness as CaCO ₃	APHA, 2340 C, EDTA Titrimetric Method	396.00	mg/l
5.	Calcium as Ca	APHA, 3500 Ca B, EDTA Titrimetric Method	115.20	mg/l
6.	Alkalinity as CaCO ₃	APHA, 2320 B, Titration Method	380.34	mg/l
7.	Chloride as Cl	APHA, 4500 Cl B, Argentometric Method	95.13	mg/l
8.	Magnesium as Mg	APHA, 3500 Mg B, Calculation Method	26.38	mg/l
9.	Total Suspended solids	APHA, 2540 C, Gravimetric Method	< 5.0	mg/l
10.	Sulphate as SO ₄	APHA, 4500 E, Turbidimetric Method	76.15	mg/l
11.	Fluoride as F	APHA, 4500 F D Spands Method	0.51	mg/l
12.	BOD (3 Days at 27°C)	APHA, 5210 C/ IS: 3025 (Part-44)	<5.0	mg/l
13.	COD	APHA, 5220 B, Open Reflux Method	<5.0	mg/l
14.	Nitrate as NO ₃	IS:3025 (Part-34), Chromotropic Method	1.52	mg/l
15.	Iron as Fe	APHA, 3500 Fe B 1,10 Phenanthroline Method	0.25	mg/l
16.	Copper as Cu	APHA, 3111 B, Direct Acetylene Flame Method	0.16	mg/l
17.	Lead as Pb	APHA, 3111 B, Direct Acetylene Flame Method	*BDL (**DL 0.01 mg/l)	mg/l
18.	Oil & Grease	APHA, 5520 B, Partition Gravimetric Method	<1.0	mg/l

Remark:- This water is suitable for non potable purpose. *BDL- Below Detection Limit, **DL- Detection Limit

MEENU KAUSHIK

ANALYST

(Checked By)

ADARSH KAWAT

(Checked By)

(Approved By)



VEL/E/N/TR/PN1452

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Test Report

Sample Number:	VEL/HCL/W/12	Report No.:	VEL/W/2106/24/012
Name & Address of the Party:	M/s Hindustan Copper Limited, Indian Copper Complex, Ghatsila, Disst.- Singhbhum, Jharkhand India.	Format No.:	7.8 F-01
Sample Description:	Recycle Water Sample	Party Reference No.:	NIL
Sampling Location:	Storage Tank - II (Surda Mines)	Reporting Date:	29/06/2021
Sample Collected By:	VEL Representative (Mr. Kamal)	Period of Analysis:	24/06/2021 to 29/06/2021
Sampling & Analysis Protocol:	IS: 10500-2012 & APHA	Receipt Date:	24/06/2021
Parameter Required:	As Per Work Order	Sampling Date:	21/06/2021
		Sampling Quantity:	2.0 Ltr.
		Sampling Type:	Grab
		Preservation:	Refrigerated

S. No.	Parameter	Test Method	Result	Unit
1.	pH (at 25°C)	APHA, 4500 H ⁺ B Electrometric Method	8.46	--
2.	Colour	APHA, 2120 B, Visual Comparison Method	*BDL (**DL 5Hazen)	Hazen
3.	Turbidity	APHA, 2130 B, Nephelometric Method	1.0	NTU
4.	Total Hardness as CaCO ₃	APHA, 2340 C, EDTA Titrimetric Method	463.00	mg/l
5.	Calcium as Ca	APHA, 3500 Ca B, EDTA Titrimetric Method	131.51	mg/l
6.	Alkalinity as CaCO ₃	APHA, 2320 B, Titration Method	452.44	mg/l
7.	Chloride as Cl	APHA, 4500 Cl B, Argentometric Method	109.23	mg/l
8.	Magnesium as Mg	APHA, 3500 Mg B, Calculation Method	32.77	mg/l
9.	Total Suspended solids	APHA, 2540 C, Gravimetric Method	<5.0	mg/l
10.	Sulphate as SO ₄	APHA, 4500 E, Turbidimetric Method	68.14	mg/l
11.	Fluoride as F	APHA, 4500 F D Spands Method	0.59	mg/l
12.	BOD (3 Days at 27°C)	APHA, 5210 C/ IS: 3025 (Part-44)	<5.0	mg/l
13.	COD	APHA, 5220 B, Open Reflux Method	<5.0	mg/l
14.	Nitrate as NO ₃	IS:3025 (Part-34), Chromotropic Method	0.88	mg/l
15.	Iron as Fe	APHA, 3500 Fe B 1,10 Phenanthroline Method	0.36	mg/l
16.	Copper as Cu	APHA, 3111 B, Direct Acetylene Flame Method	0.21	mg/l
17.	Lead as Pb	APHA, 3111 B, Direct Acetylene Flame Method	*BDL (**DL 0.01 mg/l)	mg/l
18.	Oil & Grease	APHA, 5520 B, Partition Gravimetric Method	<1.0	mg/l

Remark:- This water is suitable for non potable purpose. *BDL- Below Detection Limit, **DL- Detection Limit

MEENU KAUSHIK
SR ANALYST
(Tested By)

ARJUN RAWAT
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